JAMES RICHARDSON

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PROFESSIONAL SUMMARY

Recent graduate with a strong foundation in machine learning and data science, currently applying Python and TensorFlow to real-world problems in the Cambridge tech scene. I spent my first year after university building recommendation systems and working with computer vision datasets, learning how to translate academic theory into production code. I'm particularly interested in working with structured data and improving model performance through careful feature engineering rather than just throwing more parameters at problems.

WORK EXPERIENCE

Machine Learning Engineer

DataVault Solutions

Cambridge, United Kingdom

datavalult.io

June 2023 - Present

Developed and deployed a customer churn prediction model using scikit-learn and XGBoost that identified at-risk accounts with 87% accuracy, resulting in targeted retention campaigns for the sales team. Preprocessed and cleaned datasets containing over 500,000 customer records, handling missing values and feature scaling to prepare data for model training. Built an automated pipeline using Python and Airflow to retrain the churn model weekly on fresh data, reducing manual intervention and ensuring predictions remained current. Documented model assumptions and limitations in technical reports for stakeholders, helping non-technical team members understand when and how to trust the model's outputs.

Junior Data Scientist

Lighthouse Analytics Ltd

Cambridge, United Kingdom

lighthouseanalytics.co

January 2023 - May 2023

Collaborated with the computer vision team to label and prepare 15,000 images for an object detection project, implementing quality control checks to ensure consistent annotation standards. Trained multiple convolutional neural networks using TensorFlow and Keras, experimenting with different architectures to improve inference speed on edge devices. Created visualizations of model predictions using matplotlib and plotly to help product managers understand what the model was learning from the data. Participated in weekly code reviews with senior engineers, receiving feedback that improved my ability to write cleaner and more maintainable Python code.

Data Analyst Intern

Cambridge Insights Ltd

Cambridge, United Kingdom

cambridgeinsights.net

September 2022 - December 2022

Performed exploratory data analysis on marketing campaign datasets using pandas and SQL, identifying trends that informed decisions about budget allocation across channels. Built a dashboard in Tableau to track key

performance indicators for the marketing team, automating manual reporting that previously took several hours each week. Wrote SQL queries to extract customer behavior data from the company database, ensuring accuracy through validation checks against source systems. Assisted senior analysts with statistical testing on A/B test results, learning how to apply hypothesis testing to real business questions.

EDUCATION

University of Cambridge

Bachelor of Arts in Mathematics with specialization in Statistics

Cambridge, United Kingdom

Graduated June 2022

SKILLS

Programming Languages: Python, SQL, R

Machine Learning: Scikit-learn, TensorFlow, Keras, XGBoost, Neural Networks, Computer Vision, Feature Engineering

Data Processing: Pandas, NumPy, Data Cleaning, Exploratory Data Analysis

Visualization: Matplotlib, Plotly, Tableau

Tools and Platforms: Jupyter Notebook, Git, Airflow, Linux Command Line

Statistical Methods: Hypothesis Testing, Regression Analysis, Classification

Database: PostgreSQL, MySQL